

PROJECT 1: DATA CLEANING & PREPARATION

By

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Project Overview & Objective

As part of the Decode Labs data analytics training program, Project 1 focuses on transforming raw, unstructured dataset anomalies into a clean, standardized format. The objective is to systematically handle missing values in string/categorical fields, eliminate duplicate order IDs, and standardize string formatting and date/numeric precision.

Data Cleaning Steps

Step 1: Handling Missing Values (Categorical Imputation)

I checked for missing values across all columns using `df.isnull().sum()`. Identified 309 missing values in `CouponCode`. I also imputed all missing `CouponCode` entries with `'NONE'` using `.fillna('NONE')`.

Step 2: Duplicate Detection & Removal

For my second step, I checked for duplicate `OrderID` entries (`df['OrderID'].duplicated().sum()`) and full row duplicates (`df.duplicated().sum()`). Deduplicated records based on `OrderID` using `df.drop_duplicates(subset=['OrderID'], keep='first')`.

Step 3: Text Standardization & Whitespace Trimming

I selected all object/string columns (`df.select_dtypes(include=['object']).columns`) and applied `.str.strip().str.title()` to trim leading/trailing whitespace and convert text to Title Case. Standardized `'None'` text values back to upper case `'NONE'` for `CouponCode`.

Step 4: Date Formatting & Numeric Precision

Converted the `Date` column to standard ISO format (`YYYY-MM-DD`) using `pd.to_datetime(df['Date']).dt.strftime('%Y-%m-%d')`. Rounded numeric currency columns (`UnitPrice` and `TotalPrice`) to 2 decimal places using `.round(2)`.

3. Exact Audit Change Log

The change log below strictly reflects the code operations performed in the Python notebook:

Change ID	Code Action / Description	Target Column	Status
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CR001	Imputed 309 missing values with 'NONE' (<code>`fillna('NONE')`</code>)	CouponCode	Resolved
CR002	Deduplicated rows keeping first occurrence (<code>`drop_duplicates`</code>)	OrderID	Resolved
CR003	Trimmed whitespace and capitalized words (<code>`str.strip().str.title()`</code>)	All String Columns	Resolved
CR004	Standardized 'None' strings back to uppercase 'NONE'	CouponCode	Resolved
CR005	Formatted dates to ISO standard YYYY-MM-DD (<code>`strftime('%Y-%m-%d')`</code>)	Date	Resolved
CR006	Rounded numerical values to 2 decimal places (<code>`round(2)`</code>)	UnitPrice, TotalPrice	Resolved

4. Final Data Verification Output

As printed in the final cell of the Python notebook:

- Total Missing Values Remaining: 0
- Duplicate OrderIDs Remaining: 0
- Final Cleaned Dataset Shape: (1,200 rows, 14 columns)
- Cleaned file exported as: Cleaned_Dataset_Data_Analytics.xlsx